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# H2O HYDROGEN TORCH

## STARTER COURSE

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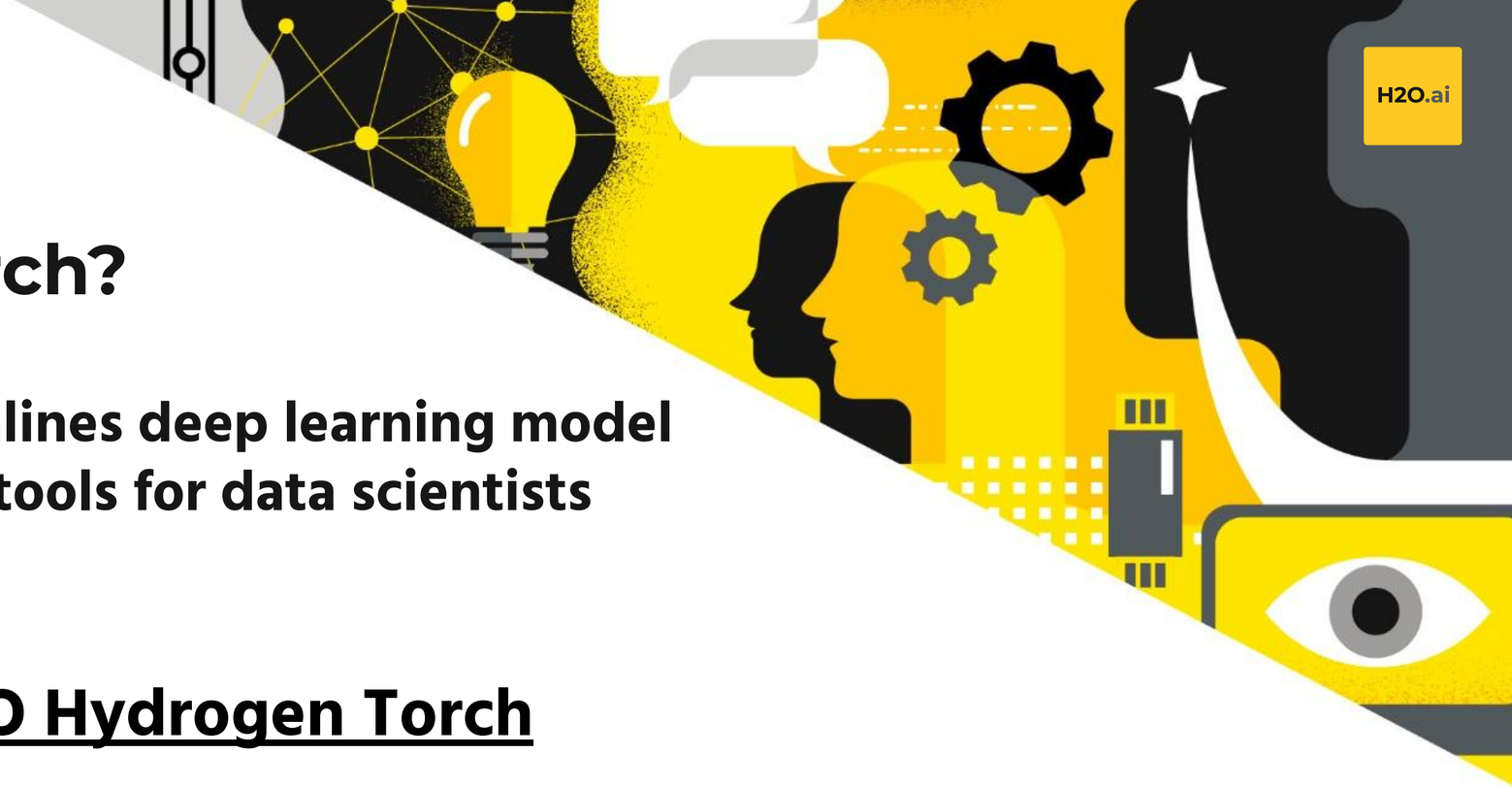



# What is Hydrogen Torch?

➔ **H2O Hydrogen Torch streamlines deep learning model training and provides extra tools for data scientists during model development.**

## Benefits of H2O Hydrogen Torch

- Hyperparameter tuning

  H2O Hydrogen Torch lets you customize hyperparameters for optimal deep learning models with top-tier performance.

- User-friendly interface with interactive charts

- H2O Hydrogen Torch offers a user-friendly interface with interactive charts for clear hyperparameter impact visualization.

- Flexible model deployment

  H2O Hydrogen Torch provides flexible model deployment options, allowing deployment within the app, in external Python environments, or via the H2O MLOps interface.

# H20 Hydrogen Torch Use Cases

- **Image**

  Classify, detect, segment, and search images, such as identifying pneumonia on chest X-rays, detecting vehicles in traffic, and finding similar products.

- **3D Image**

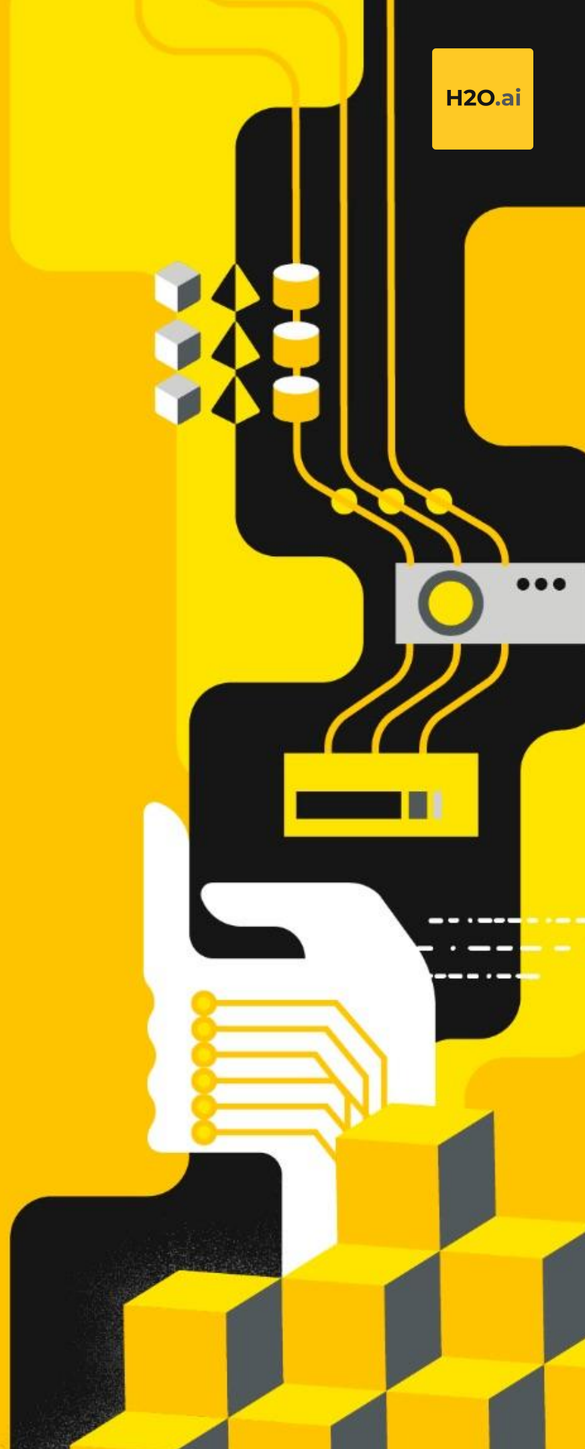
  Classify and segment 3D images, such as identifying lesion types in lungs or brain MRI images.

- **Text**

  Classify, extract, and generate text, such as predicting customer satisfaction from transcribed phone calls, extracting named entities like names and locations from medical text, and generating summaries of long documents.

- **Audio**

  Classify and transcribe audio, such as detecting bird and frog species from tropical audio recordings and transcribing call center recordings.



# H2O Hydrogen Torch Model Evaluation Scores

- **Validation Score**: Aim for a validation score of 0, which indicates perfect predictions.
- **Loss**: It measures the penalty for a bad prediction. A loss value of 0 indicates a perfect prediction.
- **MAE**: The Mean Absolute Error (MAE) is the default scorer for H2O Hydrogen Torch models. It measures the average absolute difference between the model's predicted sums and the true sums. A lower MAE value indicates better predictions.

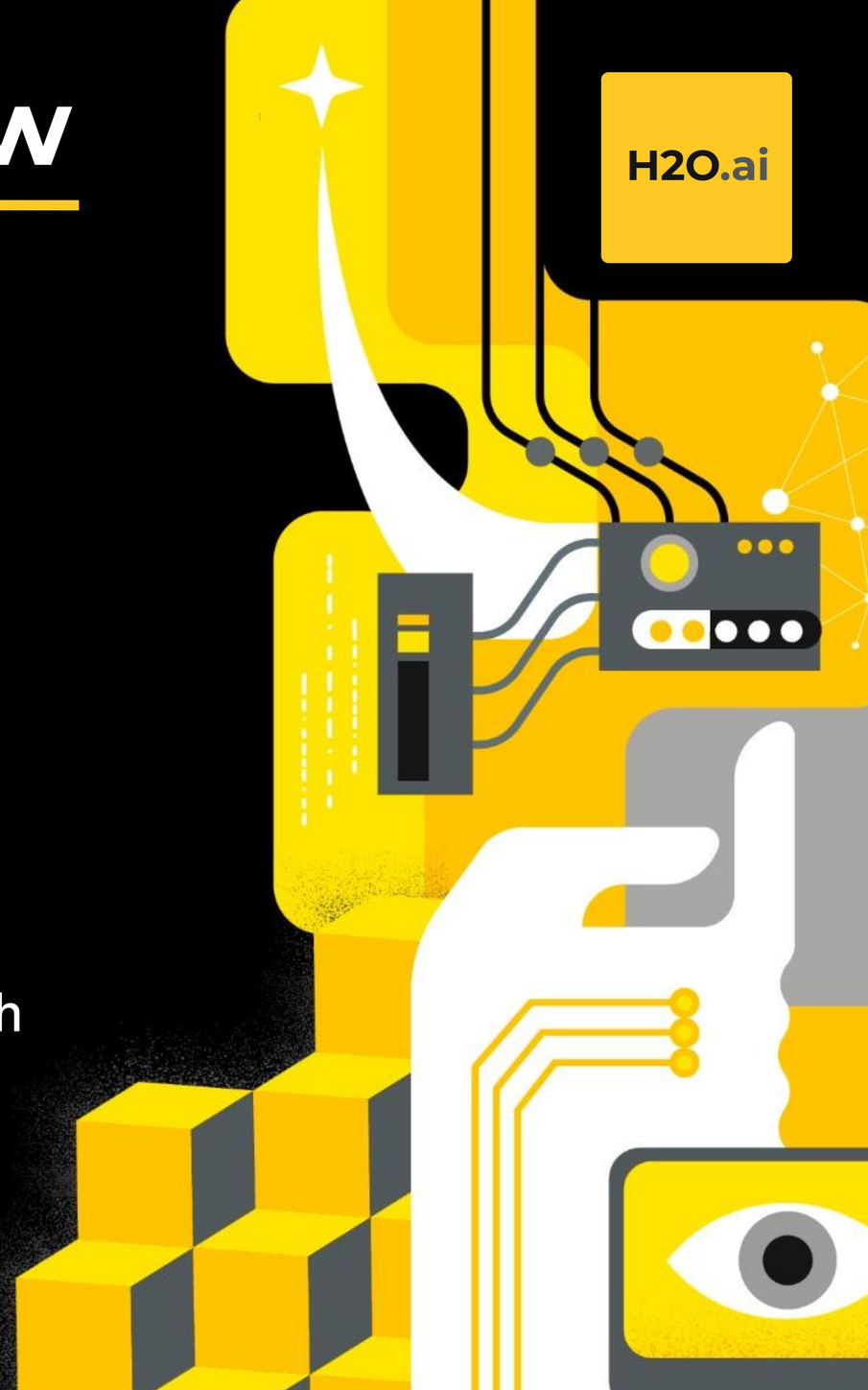
***Tracking Experiments:***
*Monitor your experiments in the View experiments card to track progress and identify any issues.*

# Reviewing Experiment Flow

- Dataset Availability
  We have access to the properly formatted dataset.

- Model Construction
  Let's build the model using H2O Hydrogen Torch. We will train, observe, and inspect the completed model.

- Model Tuning
  We will examine the hyperparameters and explore grid search techniques to optimize the model's performance.



# Hydrogen Torch Workflow

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- ***Step 1: Dataset Import***

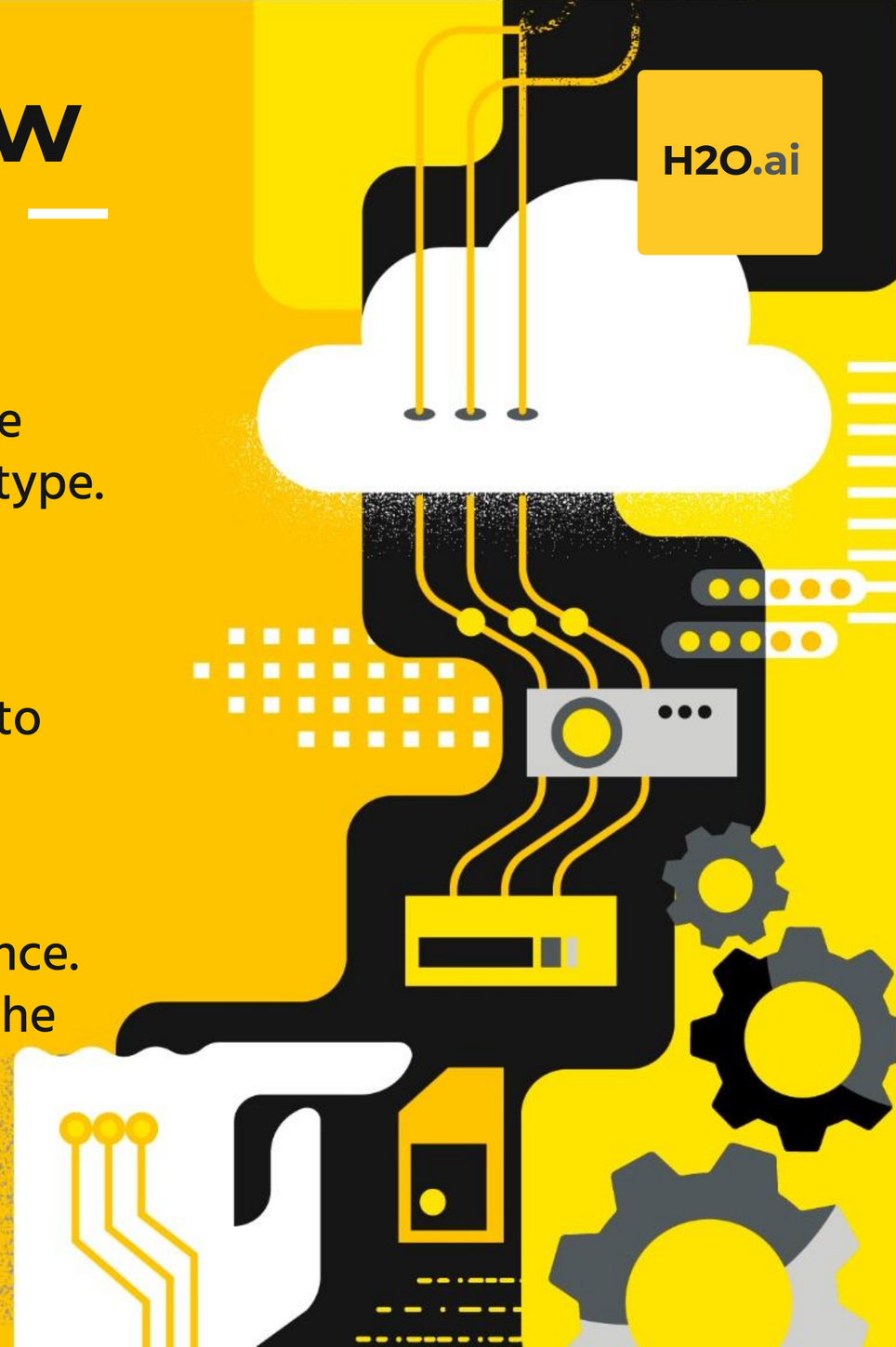
  Import your dataset into H2O Hydrogen Torch. Ensure that the dataset is in the required format, depending on the problem type.

- ***Step 2: Model Training***

  Train your model using the imported dataset. Adjust the hyperparameters as needed. You can also enable grid search to experiment with multiple hyperparameter values.

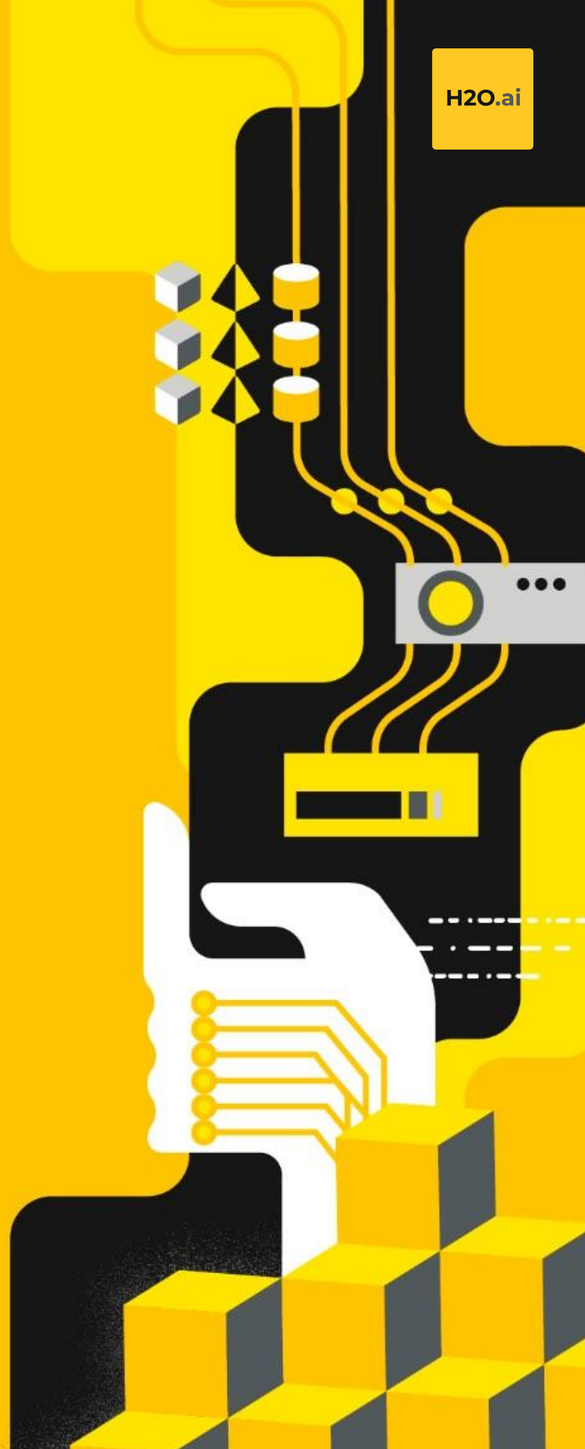
- ***Step 3: Model Inspection and Deployment***

  Once the model is trained, analyze and evaluate its performance. Use H2O Hydrogen Torch's interactive graphs to understand the impact of hyperparameters. Finally, deploy your model for practical use.



H2O Hydrogen Torch offers various deployment options, including:

- UI integration
- external Python environments and
- H2O MLOps



H2O Hydrogen Torch offers a self-contained Python scoring pipeline for Linux systems.

# H2O Hydrogen Torch Model Evaluation

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## Tips

- ★ Use a separate validation set to evaluate your model. This will help to ensure that your model is not overfitting the training data.
- ★ Track the validation score and MAE over time. This will help you to identify any signs of overfitting or underfitting.
- ★ If you are not satisfied with the performance of your model, try adjusting the hyperparameters or using a different model architecture.
- ★ Experiment with different features and preprocessing steps to improve the performance of your model.

| Step | Description |
|---|---|
| 1 | Objective Definition: Create an image regression model using H2O Hydrogen Torch for R$ coin prediction. |
| 2 | Default Hyperparameters: Utilize expertly optimized default hyperparameter values from Kaggle best practices. |
| 3 | Training and Monitoring: Observe the training process and analyze the effects of chosen hyperparameter values. |
| 4 | Interactive Charts: Utilize intuitive charts to gain insights into the impact of hyperparameters on model training. |
| 5 | Enhanced Understanding: Through monitoring and chart analysis, improve understanding of model training dynamics. |

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# H2O Hydrogen Torch

v1.3.0

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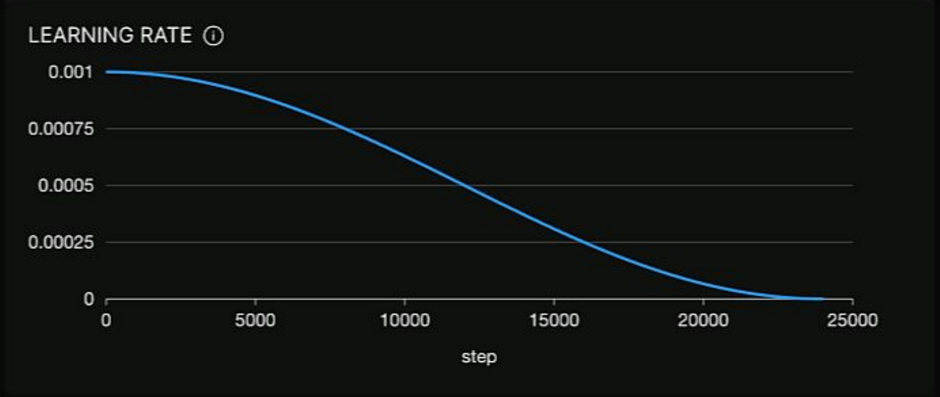
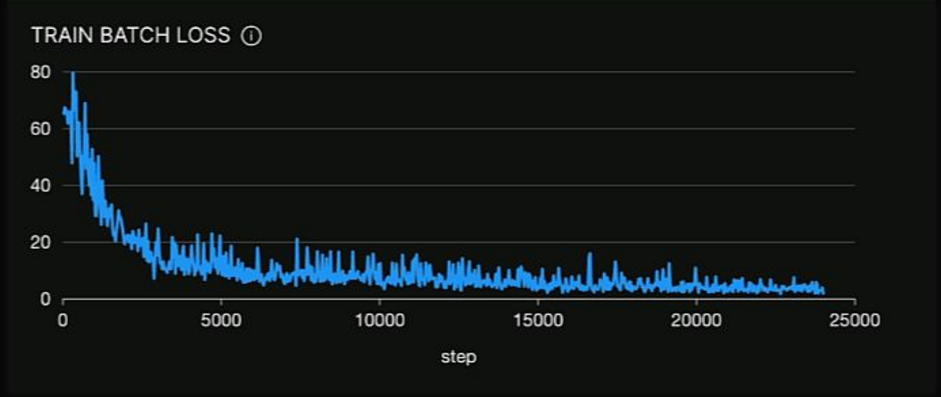
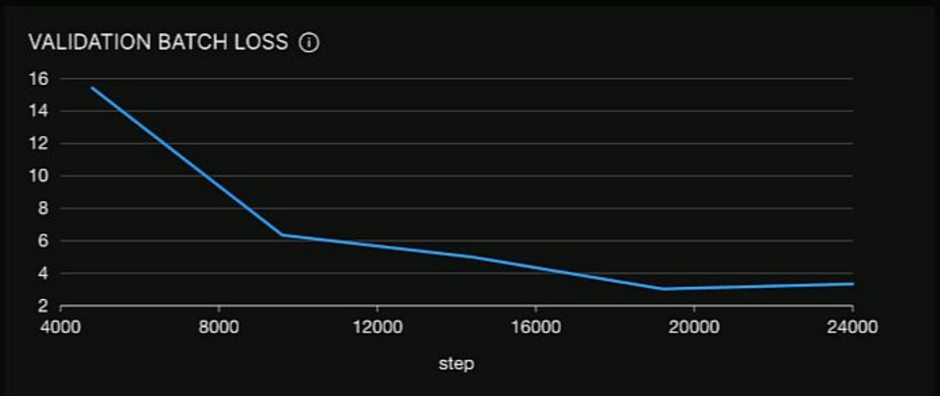
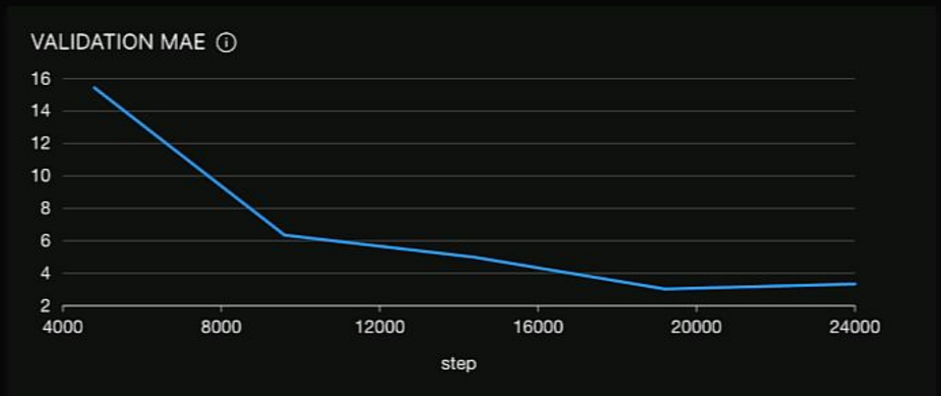
Charts | Summary | Metrics | Train Data Insights | Validation Prediction Insights | Validation Interpretation Insights | Logs | Config | Deployments




Refresh | Predict | Download logs/config | Download predictions | Download scoring | Download MLOps | Back

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# H2O Hydrogen Torch

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Navigation

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Predictions

Charts | Summary | Metrics | Train Data Insights | **Validation Prediction Insights** | Validation Interpretation Insights | Logs | Config | Deployments

Visualization of validation samples and their predictions. Hover for label information.

Random validation samples | Best validation samples | Worst validation samples

Loss: 0.000

Loss: 0.001

Loss: 0.010

Loss: 0.010

Loss: 0.011

Loss: 0.015

Loss: 0.015

Loss: 0.019

Refresh | Predict | Download logs/config | Download predictions | Download scoring | Download MLOps | Back

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By optimizing the hyperparameters, we aim to enhance the model's performance and achieve a lower validation MAE.

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The acceptability of the model in this context depends on whether the validation MAE aligns with the established margin of error.

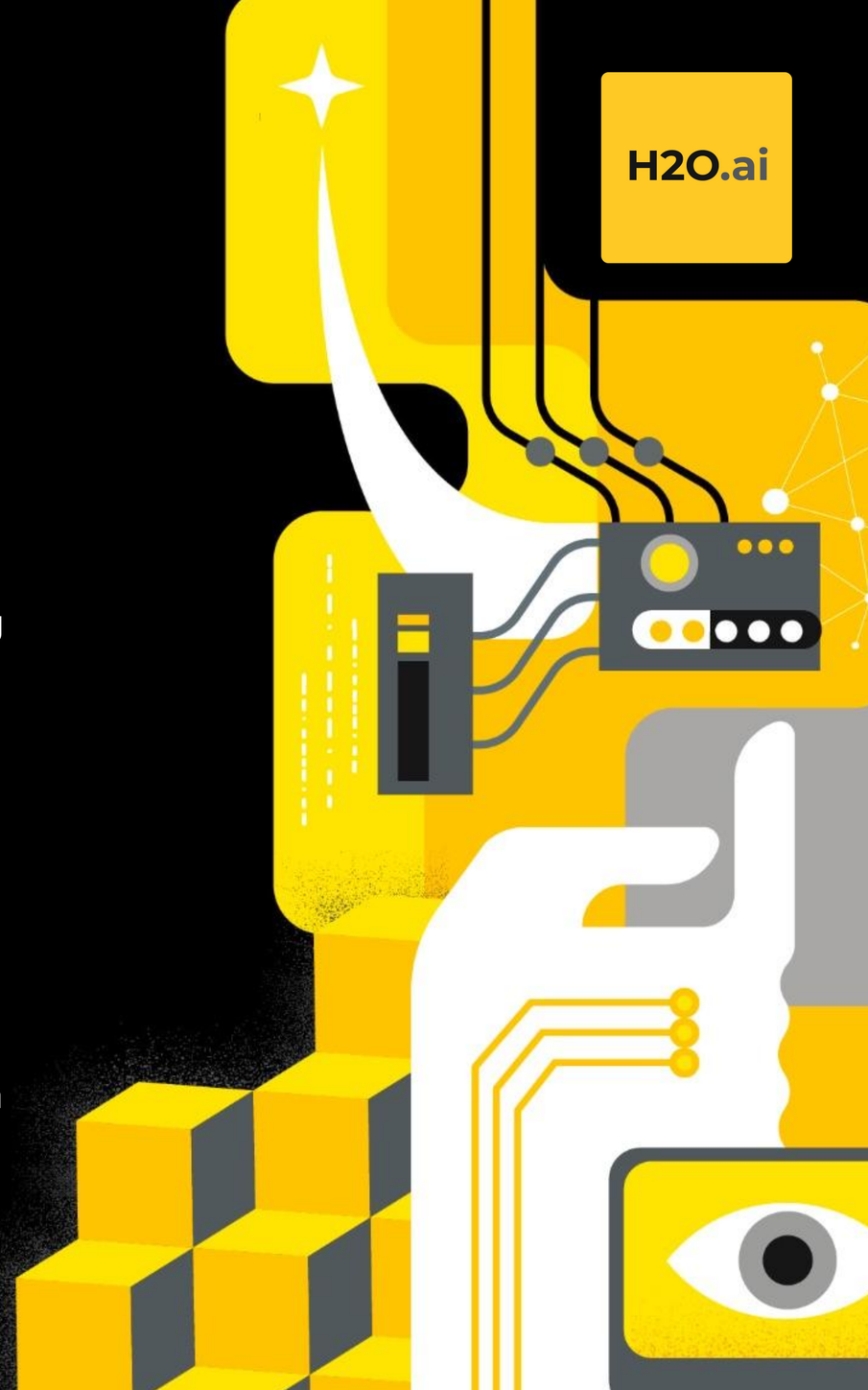
# Summary on starting a first new Experiment.

- **Fundamental concepts**: Introduction to H2O Hydrogen Torch, its purpose, application scenarios, accessibility, and experiment workflow.
- **Dataset importation and exploration**: Importing and exploring the Coins image regression dataset to familiarize oneself with its format and content.
- **Model construction and training**: Constructing an image regression model using default hyperparameter values and running the experiment, closely monitoring its progress using interactive charts and metrics.
- **Model evaluation**: Assessing the performance of the trained model by focusing on the validation MAE (Mean Absolute Error) score and analyzing both the best and worst validation samples.
- **Model acceptability**: Discussing the acceptability of the model in different use cases, such as ATM coin counting and rapid approximate coin sum estimation.
- **Model improvement**: Introducing the subsequent modules that will delve into the process of improving the model using grid search.
- **Model deployment**: Discussing the three deployment options available in H2O Hydrogen Torch:

  *-Utilizing the H2O Hydrogen Torch UI*

  *-Employing a model's Python scoring pipeline*

  *-Utilizing a model's H2O MLOps pipeline*



# H2O Hydrogen Torch Grid Search

## ❑ To enable grid search and assign multiple values to specific hyperparameters:

➢ Open the Grid search drop-down menu.
➢ Select "Custom grid."

## ❑ Note

- Custom grid: Manually select multiple values for the grid search hyperparameters.
- Grid search modes: Each mode serves a different purpose. For more information, see the "Grid search" section in the H2O Hydrogen Torch documentation.
- Default settings: H2O Hydrogen Torch displays a subset of settings for an image metric learning experiment. To display all available settings, choose "Master" in the Experience level list.

## ❑ Experience level

★ Master: Displays all available settings and allows you to customize the grid search parameters.

★ Expert: Displays a limited set of settings and does not allow you to customize the grid search parameters.

# Grid Search Optimization for Bicycle Image Metrics

- **Module Focus**: Image metric learning model for assessing similarity or dissimilarity between bicycle images.

- **Methodology**: Grid search for hyperparameter optimization.

- **Objective**: Enhance the built model's performance through systematic exploration of hyperparameter combinations.

# Tutorial Aim

1. Enhance understanding of grid search in H2O Hydrogen Torch.
2. Fine-tune and improve existing models.
3. Expedite model development by eliminating repetitive processes.

# Prerequisites:

- Basic understanding of neural network training.
- Familiarity with model training, hyperparameters, and evaluation metrics.
- Completion of the "Start Your First Experiment" modules in H2O Hydrogen Torch.

# Some Backbone Concepts

- The choice of pre-trained models used is determined by the backbone.
- The backbone is widely considered the most crucial hyperparameter in this context.
- During the fine-tuning process, we usually pinpoint promising models by adjusting the backbone.

## Hyperparameters Examples

- **Learning Rate**: A hyperparameter that determines the step size at which the model updates its weights during training.
- **Number of Hidden Layers**: The count of intermediary layers between the input and output layers in a neural network, influencing the model's capacity to learn complex patterns.
- **Regularization Strength**: A hyperparameter controlling the extent to which a model penalizes complex or large weights, helping prevent overfitting by discouraging overly intricate models.

## Grid Search

**Grid search is a hyperparameter tuning technique that systematically evaluates a predefined set of hyperparameter values to identify the combination that yields the best model performance.**

**Pros:**

- Comprehensive exploration of hyperparameter space.
- Systematic and easy to implement.

## Grid Search Modes

H2O Hydrogen Torch enables custom grid searches, allowing manual selection of multiple values for grid search hyperparameters.

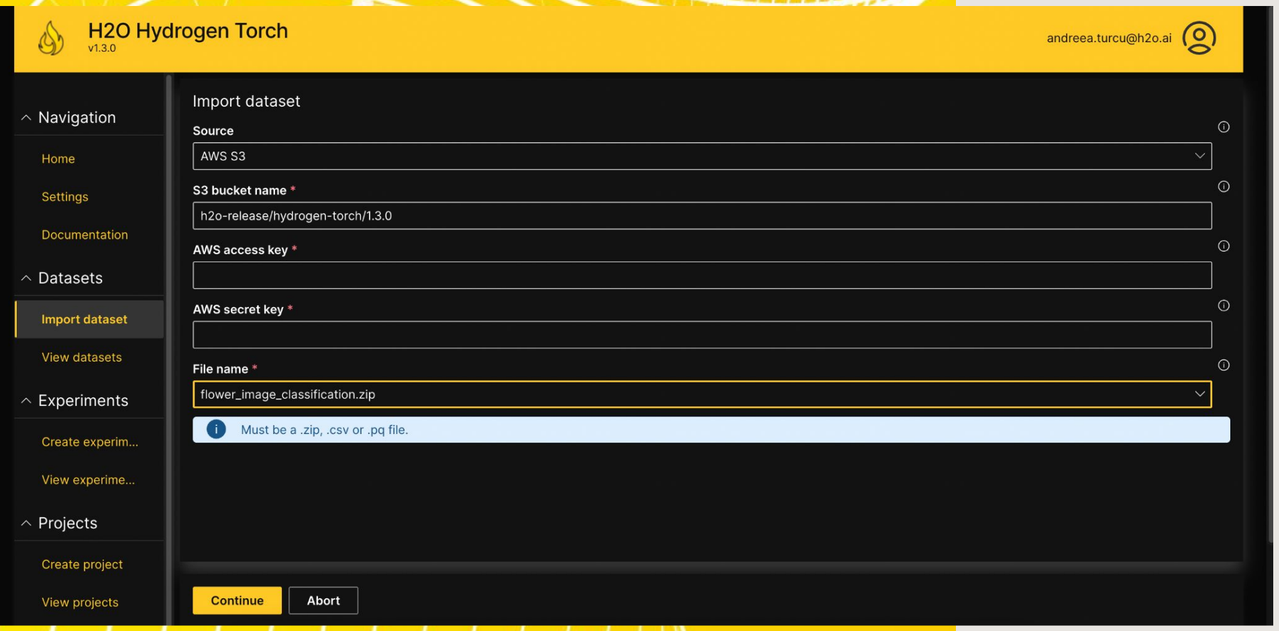
Refer to the "Grid search" section for a comprehensive understanding of these modes.

By default, H2O Hydrogen Torch shows a subset of settings for an image metric learning experiment.

# Assignment 1

## Start Your Own Experiment

*Congratulations on completing the first modules of our learning path with H2O Hydrogen Torch! You have successfully created your first experiment and learned the fundamental concepts of this powerful tool.*




For **Assignment 1**, let's build on what we've covered so far.

Your task is to start your own experiment using H2O Hydrogen Torch on a new dataset that you can find in our AWS S3 source, whose file name is: **flower_image_classification.zip**.

*You have approximately 15 to 20 minutes to complete this task.*

# Assignment 1

## Start Your Own Experiment

To ensure you achieve your goal, please adhere to the following steps:

1. **Choose a dataset**: Select the **flower_image_classification.zip** dataset.

*NOTE: Please be mindful that the flowers dataset differs from the coins dataset in terms of problem type. While the coins dataset was of regression type, the flowers dataset is of classification type.*

2. **Import the dataset**: Import the dataset into H2O Hydrogen Torch to prepare it for model training.

3. **Model construction**: Construct your model using default hyperparameter values provided by H2O Hydrogen Torch.

4. **Run the experiment**: Start the experiment and closely monitor its progress using interactive charts and metrics.

5. **Performance assessment**: After completion, assess the model's performance, focusing on the appropriate evaluation metric for your task.

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# Assignment 1

## Start Your Own Experiment

Feel free to experiment and explore the possibilities with H2O Hydrogen Torch. You can choose to work with the **Coins image regression** dataset or any other dataset of your interest.

As you progress through your experiment, think about the applicability of the model in different scenarios. How well does it perform in specific use cases? What improvements can be made to enhance its accuracy?

Remember, practice is key to mastering data science. So, embrace this assignment as an opportunity to gain hands-on experience and hone your abilities with H2O Hydrogen Torch.

*Happy experimenting!*

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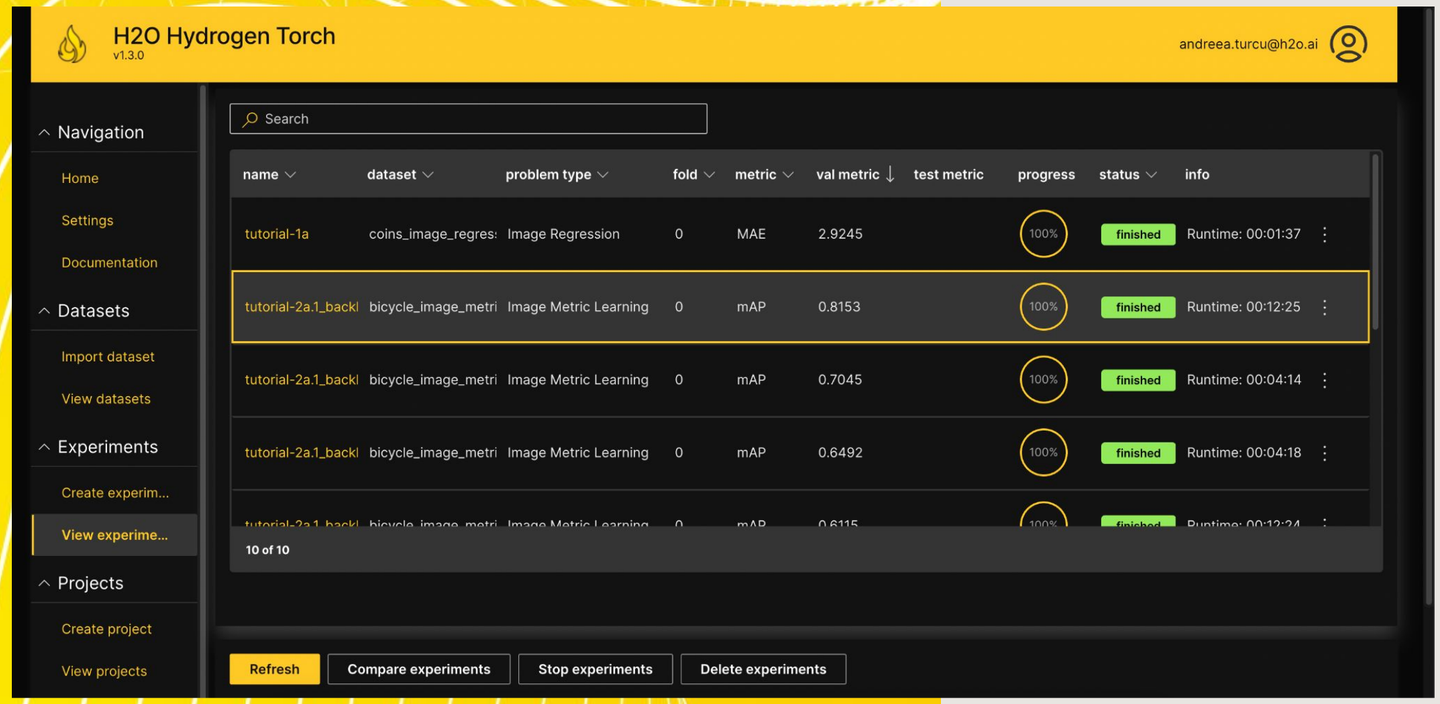
# Assignment 2

## Model Tuning with Auto Large Grid

*In this assignment, your goal is to improve your model's performance by tuning the hyperparameters of the best-performing experiment that we just created together, using the* ***bicycle_image_metric_learning.zip*** *dataset.*




Please take note that on the **View experiments** page, after all experiments have completed, you can sort them in either descending or ascending order based on the "val metric" column by simply clicking on the column name.

For sorting the table in descending order, please click on the column name twice.

# Assignment 2

## Model Tuning with Auto Large Grid

Next up please click on the 3 horizontal dots right side of the best value score and select the **Run new experiment** option.

Please select the **Auto small grid** search mode for hyperparameter optimisation and the experience level Skilled. To put it simply, the higher the level of advancement you choose, the greater the number of available hyperparameters.

For now, let's leave the experiment running, come to it later and see how these options influence the results.

Did you succeed in improving the score even more? Is there any new one that has a better score than before?

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# Assignment 3

## Beat the Best Experiment Score

*Welcome to the final assignment of the course!*

In this assignment, we'll challenge ourselves to surpass the best experiment score from **Assignment 1 - Start Your Own Experiment**, where we aimed to enhance the model's performance on the **flower_image_classification.zip** dataset.

Your objective is to follow the main steps of **Assignment 2**, but this time, we won't use the Auto Large Grid search option. Instead, we'll utilize the Custom grid search mode to personally identify the hyperparameters to tune.

For this exercise, let's select the Experience Level "Skilled" to further enhance our skills and knowledge.

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# Assignment 3

## Beat the Best Experiment Score

**Some additional information of the hyperparameters**

In the field of image classification using H2O Hydrogen Torch and deep learning parameters, there are several hyperparameters that can be fine-tuned to optimize the model's performance. Let's explore the Architecture settings hyperparameters first:

1. **Embedding Size (grid search):** This hyperparameter controls the dimensionality of numerical representations (embeddings) used to capture essential features from images.
2. **Backbone (grid search):** The backbone acts as the fundamental network responsible for extracting crucial features in image classification models.
3. **Dropout (grid search):** Dropout is a regularization technique that randomly deactivates neurons during training, preventing overreliance on specific features and aiding in generalization.

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# Assignment 3

## Beat the Best Experiment Score

Moving on to the **Training settings**, we have the following hyperparameters that can be adjusted:

- **Loss Function (grid search):** Exploring different loss functions can impact the model's performance.
- **Learning Rate (grid search):** Trying different learning rates helps find the optimal value for faster convergence and better results.
- **Batch Size (grid search):** Experimenting with different batch sizes can affect training speed and model accuracy.
- **Epochs (grid search):** Adjusting the number of epochs helps find the right balance between overfitting and underfitting.

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# Assignment 3

## Beat the Best Experiment Score

In the **Prediction settings**, we have the metric used to evaluate the model's performance during testing and validation, as well as the option of applying test time augmentations to improve the model's robustness.

Finally, in the **Environment settings**, utilizing multiple GPUs can speed up the training process and accommodate larger model architectures.

Tuning these hyperparameters will allow us to fine-tune the model and achieve improved image classification results.

After adjusting some of the model's hyperparameters, please examine and analyze the performance of your new best-rebuilt model and compare it to the initial model to measure the improvement achieved.

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# Thank you!